Align with Me, Not *TO* Me: Interactive Conceptual Alignment with LLM-Based Conversational Agents

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Abstract

Successful conversations require speakers to align on the meaning of concepts, a challenging but crucial task for human-agent communication. Large language models can generate "human-like" responses, but less is understood about 1) how they behave in the face of misaligned conceptual understanding, 2) how people perceive and respond to these behaviors, and 3) how they may affect the alignment outcome. We present results from two exploratory studies, where participants engaged in an alignment task with other people or an agent with LLM-based speech. Quantitative and qualitative analysis reveals and contextualizes potentially (un)helpful dialogue behaviors, how people perceived and adapted to the agent, as well as potential effects on alignment outcome. Our findings demonstrate the co-adaptive and co-constructive nature of conceptual alignment, necessitating a bidirectional lens. We conclude by discussing implications for agent design and future research.

CCS Concepts

• **Human-centered computing → Natural language interfaces**; *User studies*; Collaborative interaction; • **Computing methodologies → Discourse, dialogue and pragmatics**; • **Information systems** → *Language models.*

Keywords

Conceptual alignment, interactive alignment, conversational agents, human-robot dialogue, dialogue design, large language models

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1 Background

Concepts are one of the fundamental carriers of meaning in dialogue. They allow speakers to reason about categories, refer to complex ideas, and coordinate their goals and priorities. Studies of both human and human-agent dialogue have shown that people engage in *conceptual alignment*[2, 7, 17, 22], where speakers interactively build shared meaning for concepts. In human dialogues, this process is inherently bidirectional. People collaboratively refine, generalize, redefine, or even abandon the very concepts under discussion[4] to achieve conversational goals. With conversational agents, conceptual alignment serves as a foundation for alignment at a higher-level, with people's often conflicting and changing values and goals[12, 15, 16, 18]. The need for conceptual alignment is especially evident for agents situated in a shared environment, such as robots, where having an aligned representation is important for successful execution, learning, and communication in a situated environment[1, 9, 13, 14].

Large language models (LLMs) are increasingly explored to perform a variety of dialogue-based tasks that require certain levels of mutual conceptual understanding, such as learning or exchanging information and negotiating agreements[10, 11]. LLMs appear to have an innate repertoire of conceptual understanding and human-like dialogue capabilities, making it easier to implement conceptual alignment in interaction. However, recent work evaluating LLMs on general dialogue datasets have found them to lack *grounding* behaviors[20], a key mechanism to ensure what has been said is mutually understood, and required prompting informed by linguistic principles to produce the desired responses[21]. Regarding conceptual alignment, reaching a shared conceptual frame and understanding typically requires more than acts of grounding[17]. It is unclear whether LLM-based agents can readily produce the full range of required behaviors for successful conceptual alignment. Moreover, very scarce work exist to understand how people perceive these behaviors, and how different approaches to dialogue may affect the alignment outcome.

We present and discuss results from two exploratory studies with human-human and human-LLM pairs, simulating a collaborative object sorting task. Each pair sorted photos of objects based on their conceptual understanding, and engaged in spoken discussions. We compare the quantitative results of the two studies to reveal differences in linguistic behaviors and alignment outcomes. We then further contextualize the results using qualitative analysis of interviews with participants in the human-LLM study, and discuss the implications of our findings for designing dialogue to support bidirectional conceptual alignment and future research.

2 Task Design

To understand the current conceptual alignment behaviors in human and human-LLM dialogues, we design an image sorting task, inspired by the picture-naming task in psycholinguistics[3, 5]. While

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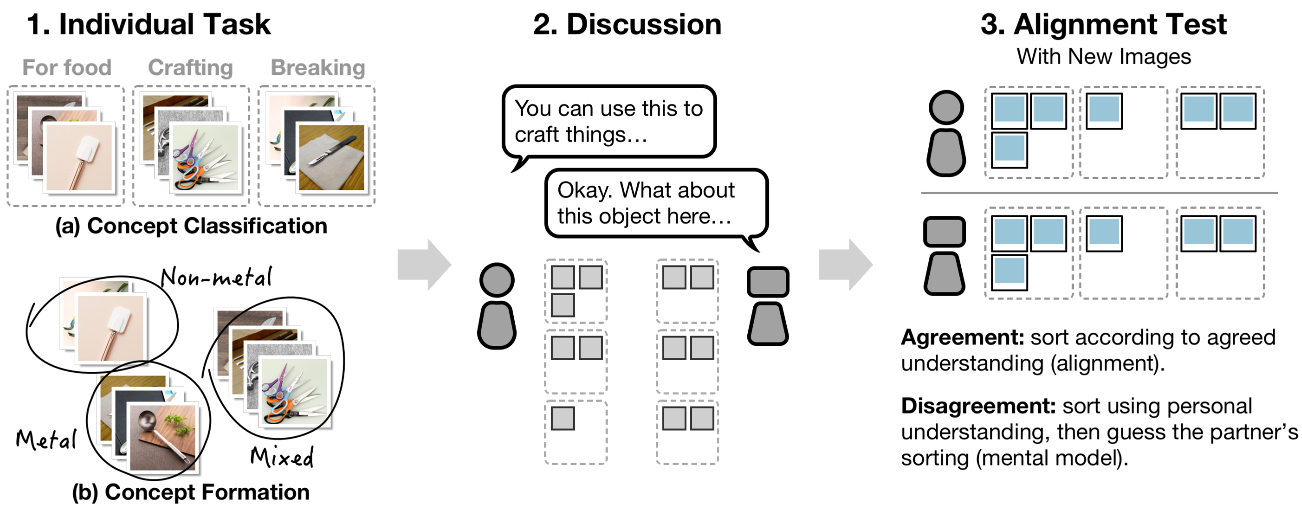

Figure 1: An overview of the concept alignment task, where a pair of participants (human-human or human-agent) sort images of objects based on their understanding of concepts. Steps 1–3 show the procedure of the task, while (a) and (b) show two forms of the individual task.

the minimal design of a conceptual alignment task could simply be asking participants to try reaching a consensus on the meaning of provided concepts through discussion, image sorting has two advantages. First, it makes explicit the differences in participants' *actual usage of concepts* and therefore may uncover misalignment where verbal discussion alone cannot. Second, it allows us to operationalize alignment as sorting similarity. In a human-agent interaction context, the objective of conceptual alignment is often to facilitate future interactions. This operationalization is closer to such an objective than, e.g., participant-reported alignment confidence.

Table 1: The concepts used for the sorting task.

Concreteness	Set	Concepts
High	A	Slab, block, stick
	B	Dark-colored, sepia-colored, multicolored
Medium	A	Container, physical support, weight
	B	For breaking, for crafting, for food
Low	A	Natural, artificial, biological
	B	Mechanical, digital, analog

As shown in Figure 1, participants first sorted images into given concepts (classification task) or grouped them to form new concepts (formation task). The concepts (Table 1) and images are curated to be ambiguous, in order to make differences in conceptual understanding more likely. They then discussed their understanding with a dialogue partner, instructed to "try to reach a consensus, but not necessarily". Finally, to test the change in alignment, we asked participants to individually sort new images according to their understanding of the concepts after the discussion. Specifically, we asked at the end of the discussion whether the pair feel like they had established a consensus, and conducted the alignment test accordingly. For those that perceived a consensus, they sorted new images according to the agreed-upon understanding this is a more direct operationalization of alignment. For those who did not, we first asked them to sort the new images according to their own understanding, then again according to their prediction of their partner's sorting. What this operationalizes is more akin to their mental model of their partner. We hypothesize that even if not consensus is reached, effective dialogue should still result in the ability to predict each other's sorting results.

3 Studies

We conducted two studies, one with pairs of human participants, one with human-LLM pairs. For the human-human study, we recruited 48 participants (29 female, 13 male, and 6 undisclosed, aged 18–40). For the human-LLM study, we recruited 12 participants (8 female, 4 male, aged 18–59) both from the participants of the previous study (N=8) and through word-of-mouth (N=4). All participants were native speakers of Mandarin Chinese.

Both the human and human-LLM studies were conducted using online meeting software. After signing a consent form, participants were asked to enter a web-based interface, and share their screen. We then introduced the tasks using an example. Each pair performed both the classification and formation task in random order, with a five-minute break in between. The discussion sessions were recorded, and participants were asked whether they felt they had established a consensus. The only difference between the human and human-LLM studies was that the latter was followed by a semi-structured interview to better understand participants' experience and thoughts on LLM-powered dialogues. As our focus is on dialogue, we decided to not include embodied interactions such as gestures and gaze, and only implemented speech interaction and vision capabilities.

The recordings were transcribed and errors manually corrected. From the human-human pairs, we recorded 2677 conversational turns from 48 dialogues (24 participants pairs, two dialogues per study session), averaging to around 57 turns per dialogue. 30 out of 48 discussions resulted in the speakers reaching a consensus. For the human-LLM pairs, we recorded 919 conversational turns from 24 dialogues, averaging to around 38 turns per dialogue. All Study 2 participants felt that they had a consensus with the LLM agent.

4 Analysis and preliminary findings

4.1 Differences in dialogue behaviors

We performed a content analysis of the transcripts of both studies and compared the distribution of the dialogue acts. We coded the transcripts according to predetermined codes. Instead of coding each uninterrupted speech segment (an "utterance") by a speaker, which may convey multiple meanings, we divided each utterance into *dialogue acts*, defined as a basic unit of dialogue that conveys one single meaning[23]. The dialogue act codes are according to their functions in dialogue.

- **Grounding** acts are for building and verifying mutual understanding. These include simple acknowledgments of understanding, indications of agreement, and asking informational or clarifying questions.
- **Argumentation** acts are related to explanation, debate and negotiation. These include presenting and attacking beliefs, requesting and providing justification, and proposing ideas or actions.
- **Social** acts are for non-task-related discussion, such as greetings and making smalltalk.

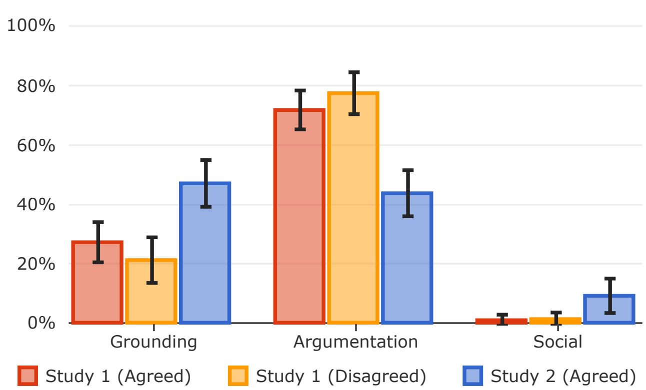

Figure 2: The distribution of dialogue acts. Agreed/disagreed is a shorthand denoting whether the participants perceived a consensus. The *Agent (Disagreed)* case have no data.

Figure 2 shows the distribution of dialogue acts from the two studies. Overall, there is a difference in distribution between Study 1 and 2. On average, people in Study 1 on average spent around 74.1% of dialogue acts to engage in argumentation, followed by 24.9% of grounding acts, and little social chatting (1.0%). On the contrary, human-agent pairs in Study 2 used around 47.1% of dialogue acts on grounding, 43.8% for argumentation, and 9.1% for social chatting. We further divided the Study 1 data into two groups according to their perceived consensus: those who felt they had reached a common, agreed-upon conception, and those who did not. The two groups showed similar distributions.

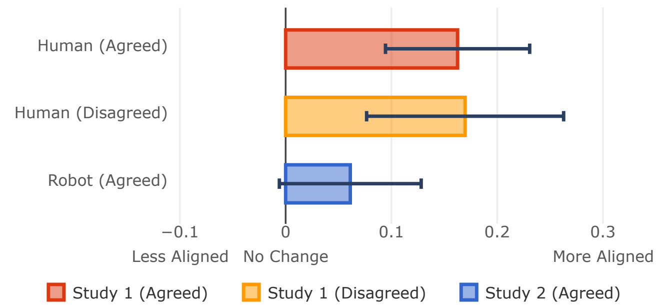

Figure 3: Changes in alignment scores from the two studies. Agreed/disagreed denotes whether the participants perceived a consensus. Error bars show 95% confidence intervals.

4.2 Changes in alignment

Figure 3 shows the changes in alignment scores from the two studies, grouped by the tasks. The alignment scores are calculated using element-centric similarity[8], which measures similarity of two clustering while taking overlaps and hierarchy into account. For participant pairs who reported having a consensus, their sorting similarity on new images showed an average change of +.163, 95% CI [+.094, +.232]. For those who did not, an average change of +.170, 95% CI [+.076, +.264]. Meanwhile, the human-LLM pairs in Study 2 had an average change of +.061, 95% CI [-.007, +.129].

4.3 Perceptions, responses, and expectations

We used thematic analysis to analyze the transcripts of the interview recordings from the human-LLM study. This process yielded high-level themes regarding people's perceptions, responses, as well as preconceptions and expectations of the LLM agent's dialogue behaviors.

Perceived behaviors. Many participants noted the LLM's **tendency to be amenable and avoid conflict** (P2, P4, P6, P7, P8, P9, P10, P11, P12). As P2 put it, "Seems that it would agree to whatever I say." The LLM was perceived to express less disagreement (P6, P7, P8, P9, P10), would not point out the participants' "mistakes"(P2, P12), and would easily retract its own belief if challenged (P7). These comments often accompanied observations on the agent's **reluctance to explain or justify** its reasoning. Participants mentioned that the agent did not explain its sorting results (P2, P6, P7, P8), and only did so when specifically asked (P6). P6 wished for a more "confrontational" discussion, believing that it would improve the understanding of both parties. Along with the tendency to agree, the agent was also perceived as **sociable**, with some participants actively chatting after the task-related conversations were over (P2, P9). Participants were positive (P2, P9) or generally neutral towards social chit-chat before and after the discussion. However, instances of chit-chat *during* task discussion were considered off-topic and frowned upon (P5, P6). The participants also pointed out a **pattern in the agent's speech** where each utterance consists of an expression of agreement, some general comments, and a final question to prompt further dialogue (e.g., "Anything else you want to discuss?") (P2, P4, P7, P11, P12). They described the questions as "mechanical

(P2, P4, P5, P7, P12)" and "too broad (P7, P9)", and considered this behavior harmful to reaching alignment (P5, P7, P8, P10).

Behaviors and confidence in alignment. Participants expressed **mixed confidence** in their alignment with the agent. While some participants expressed relatively high confidence (P2, P3, P6, P9) for both sessions, others expressed reservations about one of the sessions (P1, P4, P5, P8, P10, P12) or claimed low confidence overall (P7, P11). Each participant also named instances of agent behavior that led to their belief. **Agreeing through summarizing, paraphrasing, or extrapolating** convinced participants that their ideas had been understood (P3, P5, P10, P12). The participants expressed confidence when the agent helped summarize the discussion using simple concepts (P3) and adopted the terms they used (P12). Conversely, a plain expression of agreement or simply repeating without paraphrasing was not considered an adequate indicator of alignment (P4, P7, P8, P10). **Discussing definitions and strategies** is preferred over discussing concrete objects (P1, P3, P5, P8, P10, P11). They liked when the agent (P3, P8) or themselves (P1, P10) presented definitions for each concept and felt confident in alignment, and felt uncertain about sessions where they only discussed specific objects (P5, P10, P11). P7 also noted that people would compare not only individual definitions, but also the pros and cons of their overall strategies.

Adapting to the agent. The participants noted how they had to **adapt to a different conversation style** (P1, P11). P11 noted that the aforementioned pattern (agreement, some comments, then a question) made it more predictable, which they exploited to establish a consensus faster in the second session. Furthermore, P4 said that they became comfortable being more assertive when faced with the agent and "even to offend it". Feeling the agent lacking opposition and explanation, participants **tried guiding the agent** to produce the responses they wanted, with varied results (P3, P4, P8, P11, P12). Some participants explicitly pointed out differences to the agent to get its opinion (P11, P12), summarized their core disagreements (P12), and said things explicitly (P4) to get a satisfactory response. Meanwhile, P3 said that they couldn't find a good way to describe their ideas to the agent, and eventually gave up on discussing the point.

Preconceptions and expectations. Many participants expected the agent to be **more knowledgeable or capable than themselves** and some expressed confusion when the agent made logical or factual errors (P1, P2, P7, P8, P9) and expressed concerns (P2). However, other participants had contrasting views (P5, P6, P7, P8, P9, P11). Some equated the agent's understanding with "people's common understanding (P11)" and stressed people's ability to refuse problematic understandings (P5, P8). Some participants **expected the agent to commit to the alignment differently** (P3, P9, P10, P11). P9 and P11 believed that the agent would strictly adhere to the agreed understanding of the concepts, while they might still stick to the original understanding themselves. Many expected the agent to **act with more agency**, providing new perspectives or persuading them instead of doing as told (P5, P7, P11). Others similarly envisioned that they could talk to the agent to gain a more comprehensive understanding (P2, P12). However, while P12 believed that personalizing the agent through concept alignment is good, they stressed that the agent should still be seen as a tool rather than a person.

5 Discussion and conclusion

Our findings highlight conceptual alignment as an interactive and bidirectional process. Dialogue behaviors shape the final outcome not only through conveying information, but also through affecting subsequent behaviors, as well as people's evolving assessment of consensus, partner capability, and alignment confidence.

First, the results highlight how different dialogue behaviors may correlate to different alignment outcomes. The differences in dialogue behaviors and alignment outcomes between our two studies suggest a possible correlation between a lack of argumentative dialogue and subpar alignment outcome. Mismatched conceptual framing and understanding is commonplace in day-to-day human interactions. These differences are resolved in dialogue often through interactive negotiation[24] and mutual adjustment[4], which in turn requires acts of explanation, negotiation, and debate[19]. The same case could be true for agents, even though we might expect the agent to understand and utilize people's personal conceptions in concrete tasks and scenarios. Blanket agreement and acceptance may mask underlying misalignment, which, in turn, could limit how much information can be conveyed in a dialogue, as well as skew people's perception of their current alignment. Overly amenable behaviors may also have indirectly limited conversation depth through negative experience. Participants in our study typically framed such behaviors in negative light, and noted cases where model (in)capability/hallucinations mismatched their preconceptions. These may have led people to disengage early, as hinted by the slightly less number of turns for Study 2 (38 vs. 57 for Study 1). These potential explanations need to be be further tested in future work to establish a causal relationship.

Second, the results show how both parties adapted to each other's behavior in interaction, which collectively shaped their dialogue and the alignment outcome. In Study 2, when the agent appeared amenable, participants adapted to the agent's behaviors by becoming more assertive. When the agent focused on concrete object examples rather than definitions, participants followed, which some associated with lower confidence in alignment. Likewise, when the agent engaged in less argumentation, several described the interaction as lacking depth, and some tried guiding the agent into producing the desired output. It is notable that not all guiding attempts succeeded. The overall quantitative results still show a comparatively lower proportion of argumentation dialogue. This suggests the importance of taking into account the bidirectional nature of alignment when designing human-AI interactions. Our interviews also show that people's preconceptions of an LLM agent also shaped their behaviors and confidence, which they updated through their interaction with the agent. This could mean that simply "matching distributions", i.e., forcing the LLM to produce a similar proportion of alignment behaviors, may not automatically lead to better outcome. Instead, more needs to be understood regarding how people's preconceptions may shape their interaction, as well as how different dialogue strategies may help/impact people's understanding of agent capabilities and knowledge. Overall, these results highlight the need for a bidirectional lens, where the effects of agent dialogue design and people's contribution and understanding is best studied together.

Third, the findings draw distinctions between argumentation behaviors, perceived consensus, alignment confidence, and the alignment outcome. Human pairs in Study 1 engaged in similar proportions of argumentation regardless of whether they ended up having a consensus. Both groups had measurable improvements in either sorting similarity or how well they could predict their partner's sorting. Meanwhile, all participants in Study 2 perceived a consensus with the LLM agent with less argumentation, yet the pairs likely did not have a comparable improvement. Moreover, all participants perceived a consensus in Study 2, which did not concur with their lower confidence in alignment. Overall, these results suggest that argumentation may indicate in-depth exchange of conceptions, instead of a lack of consensus. Perceived consensus may not reliably mean higher confidence in alignment, nor an actual successful outcome.

Our work has several implications for future research. First, future work should investigate how different dialogue behaviors such as (requesting) justification, counterexamples, or explicit articulation of uncertainty may affect alignment outcomes across tasks and embodiments. Our ongoing work aims to build a design space for conceptual alignment dialogues by exploring different strategies in a concrete human-robot setting. Second, evaluation of agent/LLM alignment could benefit from incorporating interactive conceptual alignment to understand how complex concepts such as values and intentions is realized and shifts in context. To encompass the broader range of phenomenon in conceptual alignment, evaluation frameworks could incorporate separate measures of dialogue behaviors, representation shift, partner-model accuracy, as well as subjective measures of confidence and expectations. Third, technical research is needed to support the detection of conflicting or under-specified concepts in dialogue (e.g., [6]) and to develop alignment methods that balance compliant behaviors with those that promote mutual understanding (i.e., those related to explanation, negotiation, and debate). Conceptual alignment in dialogue, similar to interactive goal and value alignment at a higher level, appears to require sustained, reciprocal negotiation rather than mere acceptance or adherence. Understanding how to support such dialogues remains an open challenge for bidirectional human–AI alignment.

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